MOD-DiT rebuttal

March 2026

1 Different Method SMFS

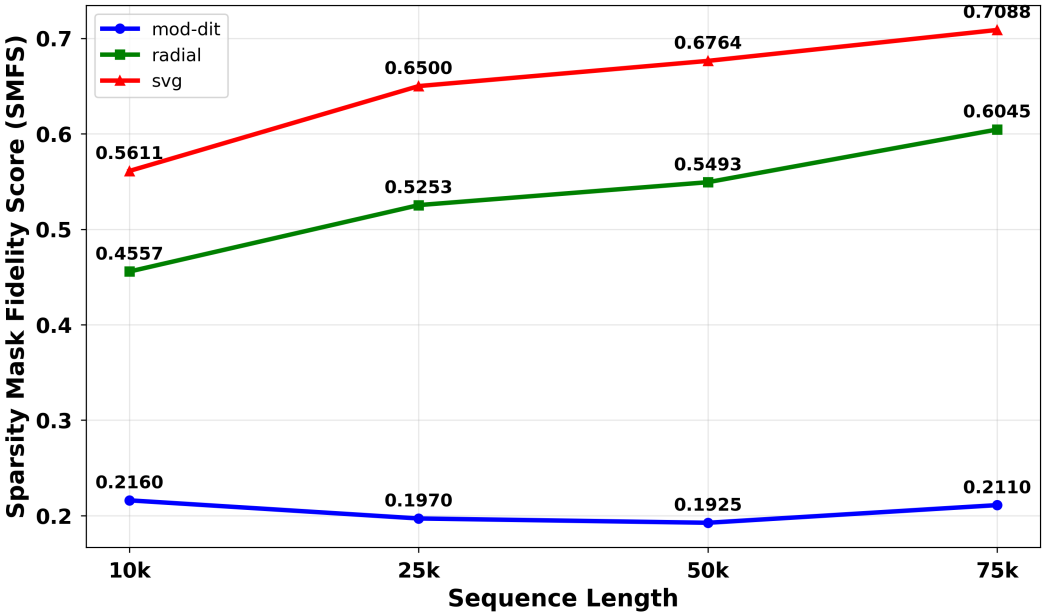

Figure 1: SMFS Comparison of Different Methods on HunyuanVideo Model. We Use 5 different prompts, randomly select 6 layers and 8 heads to calculate the average SMFS.

2 Supplementary VBench results

Abbreviations: S.C.=subject consistency, B.C.=background consistency, M.S.=motion smoothness, T.F.=temporal flickering, I.Q.=imaging quality, A.Q.=aesthetic quality, D.D.=dynamic degree

Table 1: VBench results on the Wan2.1 14B model (A100 80GB GPU, 69 frames, 768×1280 resolution)

Method	S.C.	B.C.	M.S.	T.F.	I.Q.	A.Q.	D.D.	sparsity	latency
Full	**0.9623**	**0.9655**	0.9844	**0.9789**	**0.6722**	**0.6012**	**0.9378**	0	2041s
Sparge	0.9033	0.9275	0.9798	0.9632	0.6645	0.5610	0.9043	50.10%	1389s
Radial	0.9152	0.9339	**0.9860**	0.9750	0.6620	0.5783	0.8500	59.29%	1224s
SVG	0.7986	0.8859	0.9556	0.9347	0.6133	0.5292	0.9167	73.07%	998s
Ours	0.9427	0.9488	0.9823	0.9752	0.6674	0.5827	0.9289	**79.37%**	**901s**

Table 2: VBench results on the Hunyuan model (A100 80GB GPU, 117 frames, 768×1280 resolution)

Method	S.C.	B.C.	M.S.	T.F.	I.Q.	A.Q.	D.D.	sparsity	latency
Full	**0.9582**	**0.9478**	**0.9766**	**0.9723**	**0.6693**	**0.6381**	**0.9215**	0	2370s
Sparge	0.9339	0.9156	0.9528	0.9510	0.6432	0.5603	0.9120	68%	1481.2s
Radial	0.9312	0.9290	0.9715	0.9322	0.6467	0.5715	0.8423	73.53%	1357s
SVG	0.8301	0.8711	0.9544	0.9018	0.5928	0.5330	0.9011	76.22%	1109s
Ours	0.9398	0.9320	0.9687	0.9527	0.6587	0.5899	0.9104	**80.23%**	**1044s**

3 Results under Matched Sparsity Settings

Table 3: Results under matched sparsity settings with prior methods (Wan2.1 14B model,A100 80GB GPU, 69 frames, 768×1280 resolution)

Method	S.C.	B.C.	M.S.	T.F.	I.Q.	A.Q.	D.D.	latency
Radial	0.9152	0.9339	0.9860	0.9750	0.6620	0.5783	0.8500	1224s
Ours	**0.9518**	**0.9594**	**0.9867**	**0.9776**	**0.6630**	**0.6070**	**0.9789**	**1265s**

4 Ablation Study of Block Size

Table 4: Ablation study of block size on model performance (Hunyuan model, A100 80GB GPU, 117 frames, 768×1280 resolution, aligned sparsity)

Blocksize	S.C.	B.C.	M.S.	T.F.	I.Q.	A.Q.	D.D.	latency
64	**0.9401**	**0.9338**	**0.9689**	**0.9588**	0.6451	**0.5971**	**0.9152**	1461s
128	0.9398	0.9320	0.9687	0.9527	**0.6587**	0.5899	0.9104	**1044s**

5 MOD-DiT on FastWan Distilled Model

Table 5: Results of Applying MOD-DiT to the Distilled Model FastWan (8-step)

Method	S.C.	B.C.	M.S.	T.F.	I.Q.	A.Q.	D.D.	latency
FastWan	**0.9317**	**0.9489**	**0.9766**	**0.9658**	**0.6531**	**0.5910**	**0.9201**	327s
FastWan+MOD-DiT	0.9287	0.9470	0.9692	0.9543	0.6502	0.5904	0.9176	**210s**

6 Results of MOD-DiT and SVG-2 in VBench

Table 6: VBench results on the Hunyuan model (A100 80GB GPU, 117 frames, 768×1280 resolution), SVG-2 adopts the recommended parameters from the original paper

Method	S.C.	B.C.	M.S.	T.F.	I.Q.	A.Q.	D.D.	latency
SVG-2	0.9325	**0.9369**	0.9636	0.9518	0.6431	**0.5901**	0.9076	1056s
Ours	**0.9398**	0.9320	**0.9687**	**0.9527**	**0.6587**	0.5899	**0.9104**	**1044s**

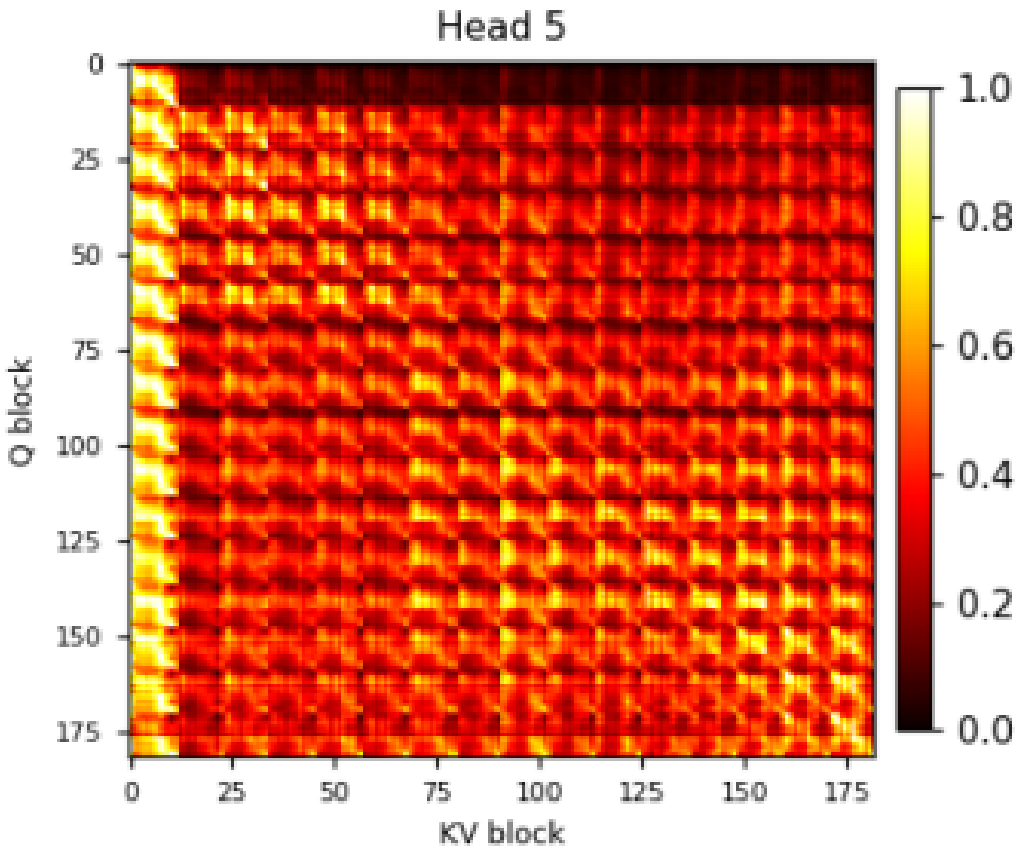

Figure 2: Attention map on FastWan Distilled Model

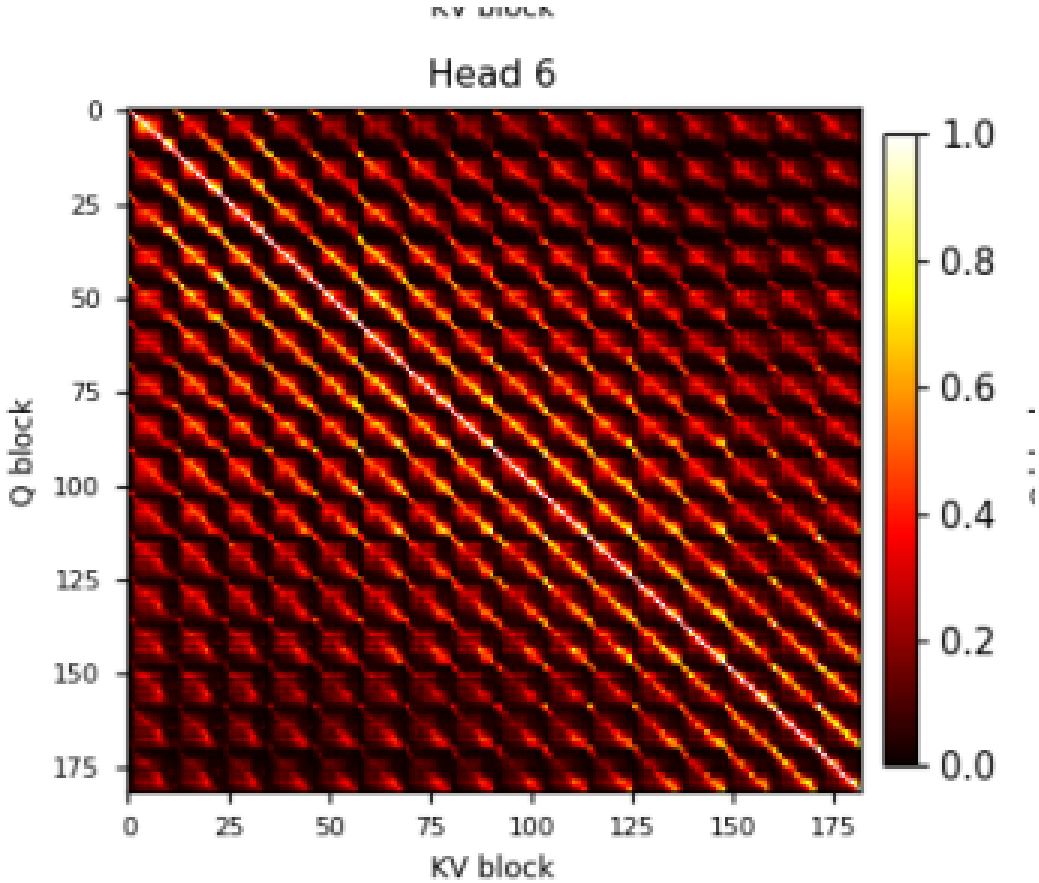

Figure 3: Attention map on FastWan Distilled Model

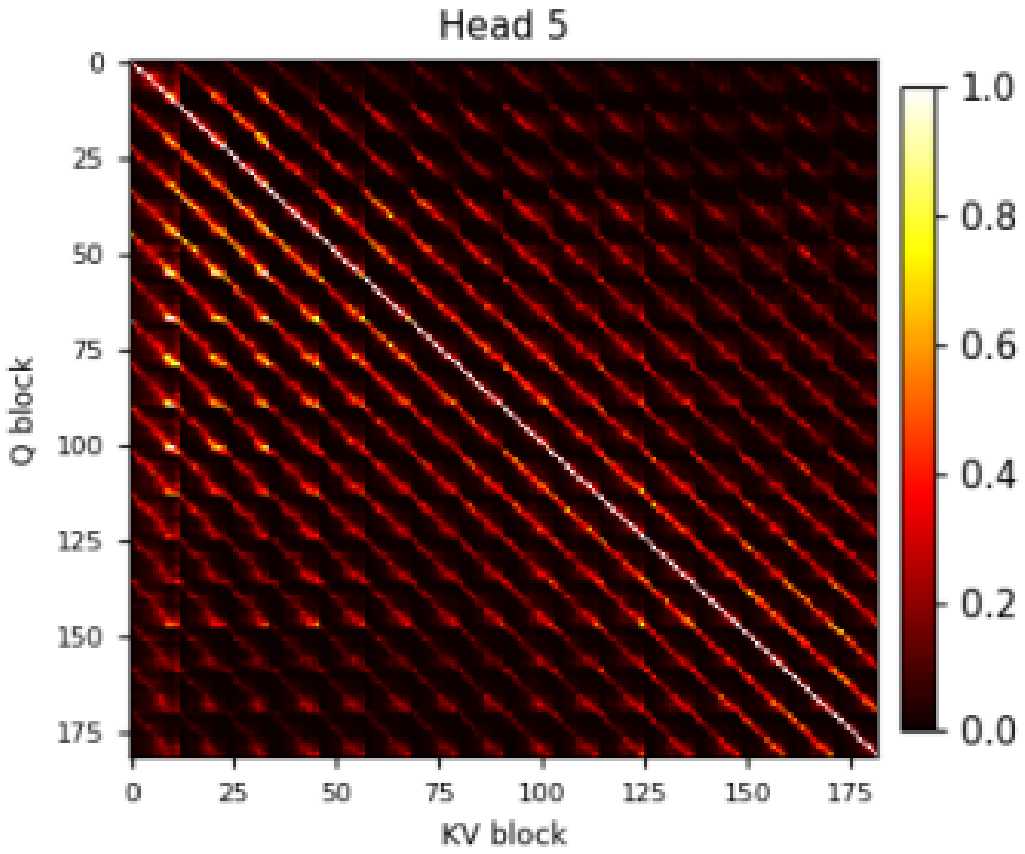

Figure 4: Attention map on FastWan Distilled Model

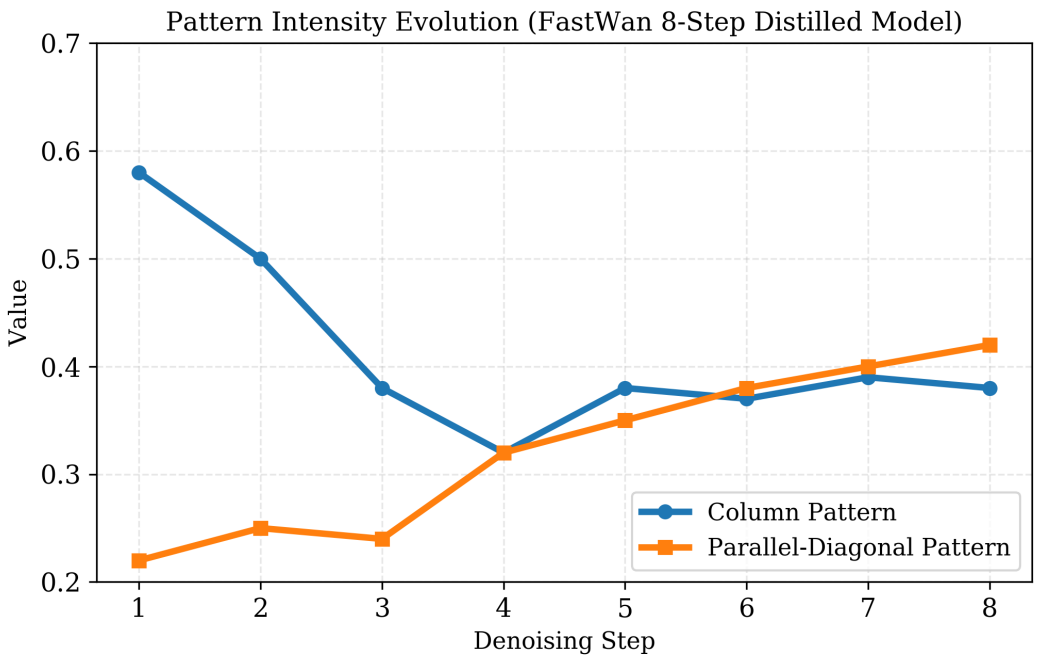

Figure 5: FastWan Distilled Model — Evolution of vertical and parallel-diagonal pattern across denoising steps

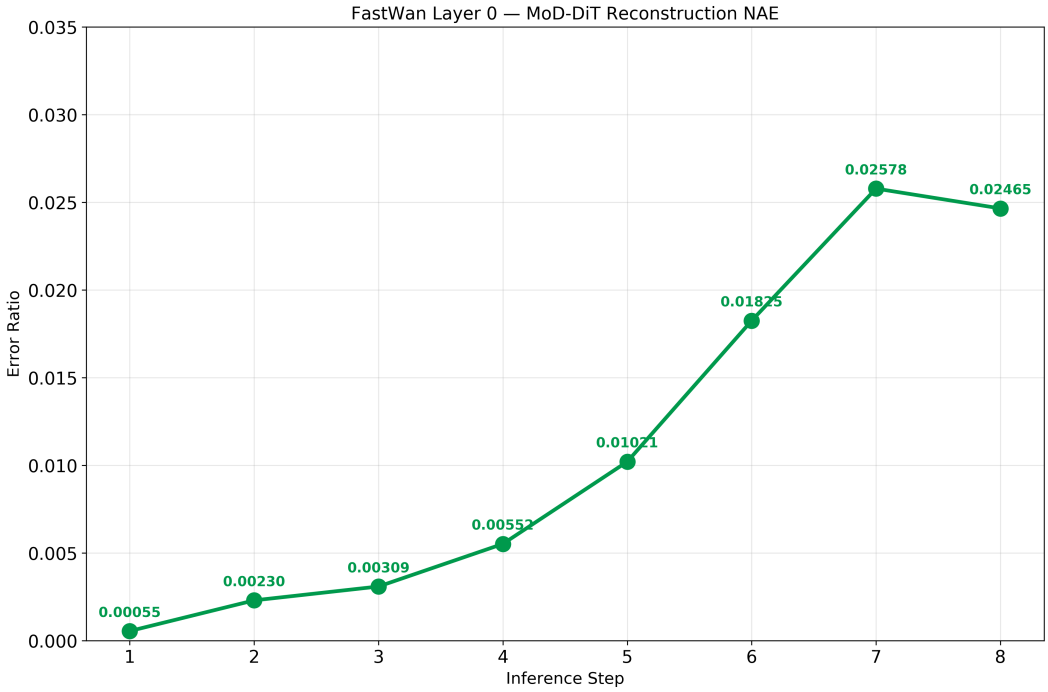

Figure 6: Normalized approximation error across denoising steps on FastWan Distilled Model

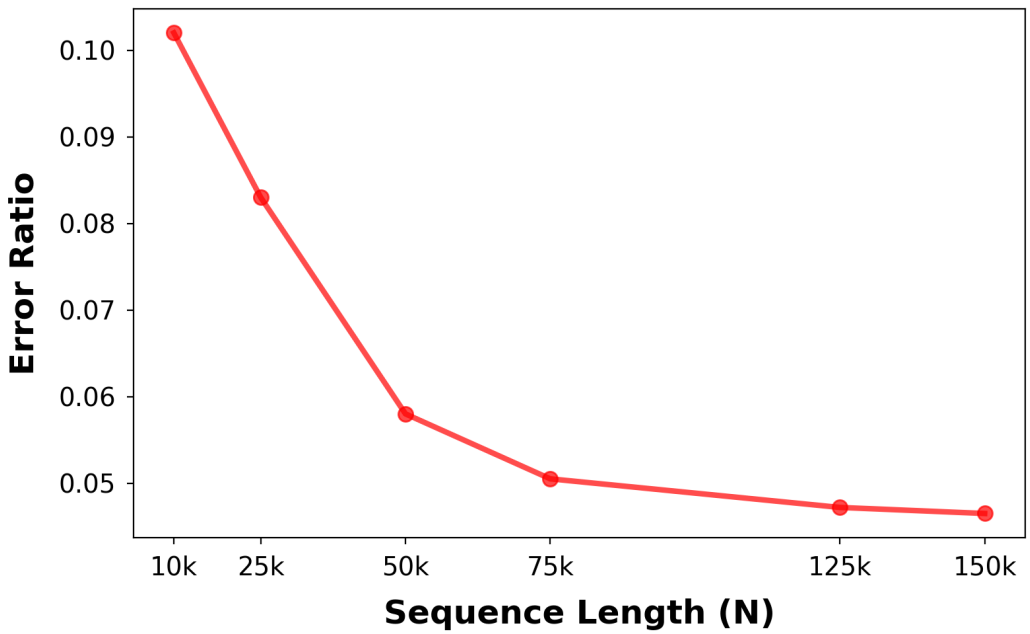

Figure 7: Normalized approximation error in long sequence (HunyuanVideo model, 720p). We use 5 different prompts, randomly select 6 layers and 8 heads to calculate the average NAE.